Verilens: A Multi-Modal AI Framework for Image Provenance and Contextual Verification

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Abstract

This paper proposes VeriLens, a multimodal AI-powered browser extension designed to combat the spread of misinformation by enabling real-time verification of images and their associated claims. As digital platforms increasingly rely on visual content, misleading reuse of authentic images in incorrect contexts has emerged as a significant driver of fake news. VeriLens integrates reverse image search with multimodal artificial intelligence to analyze both visual content and accompanying narratives. By retrieving historical instances of images and evaluating semantic consistency between the image, its context, and user-provided claims, the system generates evidence-based credibility assessments. The proposed architecture emphasizes accessibility, requiring minimal technical expertise, while maintaining privacy through transient data processing. Through pilot testing and iterative refinement, the system aims to enhance digital literacy, reduce misinformation sharing, and support critical evaluation across diverse user groups. This work highlights the potential of human-centered AI tools in addressing misinformation in the digital space.

Introduction

The rapid digitization of information has fundamentally transformed how societies consume, interpret and share news. Today, social media platforms and online news portals function as primary gateways to information for millions of individuals. While this accessibility has democratized information sharing, it has also enabled the rapid and large-scale spread of misinformation. Research published in Science by Vosoughi, Roy, and Aral demonstrates that false news spreads significantly faster and farther than truthful news online [1], and crucially, that humans rather than automated bots are primarily responsible for its circulation. This finding reframes misinformation not simply as a technological issue but as a human social challenge within digital communication networks.

The societal consequences of fake news are substantial. Research published in Information Processing & Management highlights that misinformation erodes trust in institutions, distorts democratic discourse, and contributes to political polarization and social instability [2]. The problem has intensified with the rise of generative artificial intelligence tools that are widely accessible to the public. Recent findings presented at the ACM in 2024 show that AI-generated fake news is shared at similar rates to human-made fake news [3], demonstrating that automated content creation is just as sophisticated and persuasive, as human works. This development lowers the barrier for producing large volumes of misleading content and increases the scale at which misinformation can circulate.

The urgency of addressing fake news is reflected in research published in Cognitive Computation [4], which emphasizes the growing demand for effective fake news detection systems. However, vulnerability to misinformation varies across demographic groups, and digital literacy alone does not fully mitigate risk. Studies show that older adults are more likely to encounter and share fake news compared to younger populations. Research published in a Springer chapter on misinformation indicates that fake news disproportionately reaches older people [5], and further evidence from Psychological Science in the Public Interest published by SAGE demonstrates that older adults with lower levels of digital literacy are particularly susceptible to misinformation [6]. These findings suggest that aging populations face heightened risks in digital environments where visual and textual misinformation can appear highly credible.

At the same time, younger generations are not immune to the problem. Research published through KoreaScience indicates that even young adults struggle to accurately differentiate fake news from legitimate information [7]. Similarly, a study on adolescent vulnerability to fake news and racial hoaxes reports that although younger individuals are more aware of the existence of misinformation, they still experience difficulty in reliably identifying deceptive content [8]. Awareness does not automatically translate into discernment. This highlights a broader societal need for supportive tools that enhance critical evaluation skills across all age groups.

One particularly powerful form of misinformation involves the misuse of images. Images evoke emotional responses and are often perceived as objective representations of reality. However, many instances of visual misinformation do not involve fabricated images but rather authentic images that are removed from their original context and paired with misleading captions. This is especially concerning given that a significant proportion of modern news consumption occurs in digital and visually driven formats, digital literacy must evolve to include image verification skills. To address this social problem, this proposal presents the development of an AI-powered browser extension designed to assist users in verifying images and evaluating whether associated claims are contextually accurate.

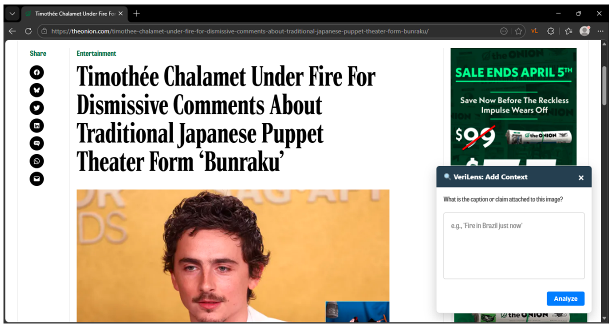

Figure 2: *The VeriLens interface allows users to enter the caption or claim associated with an image for analysis.*

Methodology and Implementation

The proposed solution is a browser extension that integrates reverse image search capabilities with multimodal artificial intelligence analysis to provide real-time contextual verification of images encountered online. The system is designed to be accessible and intuitive, requiring minimal technical knowledge from users. When browsing the internet, users can right-click on any image and select an option to verify it. The extension then prompts the user to input or confirm the caption or claim associated with the image. This ensures the AI evaluates both the image and the accompanying narrative.

Following user input, the extension performs a reverse image search using Google Lens through SearchAPI, as shown in Figure 1. The reverse search identifies prior occurrences of the image online, determines the earliest known publication date, and retrieves relevant source links. This retrieval process is essential because many misleading posts involve genuine images that are recirculated in unrelated contexts. By establishing the historical timeline and original context of an image, the system builds a factual foundation for further analysis.

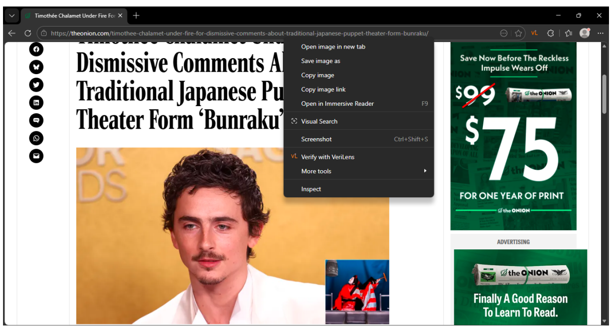

Figure 1: *VeriLens can be accessed by right-clicking an image in the browser, showing the option 'Verify with VeriLens.*

The retrieved information, along with the image and user-provided claim, is then processed through a multimodal AI model. In the prototype version, Gemini is used for this task. The AI evaluates semantic consistency between the image content, the historical context identified through reverse search, and the claim submitted by the user (Figure 2). The model assesses whether the visual evidence supports the stated claim, whether there are temporal or geographic inconsistencies, and whether the image appears to have been reused in a misleading way. The system then generates a natural language explanation that presents a credibility assessment and a justification based on retrieved evidence as demonstrated in Figure 3.

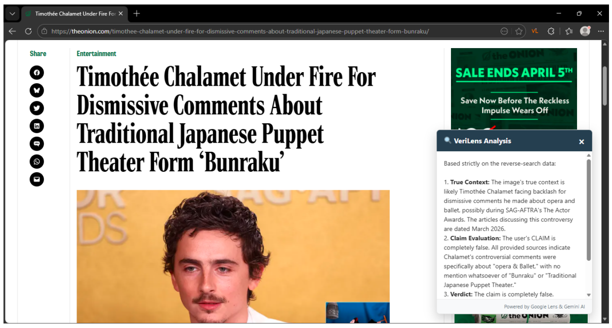

Figure 3: *VeriLens confirms the claim about Timothée Chalamet criticizing 'Bunraku' is false; the controversy was about opera and ballet, not traditional Japanese puppetry.*

The technical architecture consists of a frontend browser extension built using JavaScript that integrates with browser context menus. When activated, the extension communicates with a backend server through secure HTTPS protocols. The backend functions as an API gateway that manages requests to SearchAPI and the Gemini model. A caching mechanism stores frequently searched image metadata to reduce redundant API calls and control operational costs. Importantly, the system is designed to minimize data retention. Images are processed transiently and are not permanently stored, thereby reducing privacy risks.

Implementation is structured across three phases. The first phase focuses on refining the prototype, enhancing the user interface, improving prompt engineering for multimodal analysis, and implementing cost control mechanisms such as caching. The second phase involves pilot testing with users across age groups to evaluate usability, clarity of explanations, and detection accuracy. Feedback gathered during this stage informs model refinement and interface adjustments. The third phase focuses on scaling deployment through cloud infrastructure optimization and partnerships with educational institutions and digital literacy initiatives. By structuring development incrementally, the project balances innovation with practical validation.

Ethics & Social Impact

The social impact of this browser extension lies in its potential to directly reduce the spread of misinformation by empowering individuals with accessible verification tools. Given that humans are the primary spreaders of fake news, as demonstrated in Science, empowering users at the point of content consumption can meaningfully interrupt the misinformation cycle [1]. Older adults, who research indicates are particularly vulnerable to fake news exposure, stand to benefit significantly from a tool that provides contextual verification without requiring advanced technical skills. At the same time, younger individuals who struggle with distinguishing misinformation despite higher awareness levels may also benefit from guided evaluation prompts that reinforce critical thinking.

The extension contributes to digital literacy by embedding verification practices directly into everyday browsing behavior. Rather than expecting users to independently conduct reverse searches or consult external fact-checking websites, the system integrates verification into the browsing interface itself. Over time, this repeated interaction may strengthen users' evaluative skills and reduce impulsive sharing behaviors. Performance indicators for social impact include reductions in the sharing of flagged misinformation during pilot studies, improvements in user confidence when assessing online images, and alignment between AI assessments and professional fact-checker evaluations.

Ethical considerations are central to the deployment of such a system. Fairness requires that the AI model be evaluated across diverse datasets to minimize demographic or cultural bias in assessments. Regular audits and transparent reporting of model performance are necessary to ensure equitable treatment of different content domains. Privacy is protected by minimizing data storage and encrypting communications between the browser extension and backend server. Users are informed that the tool provides probabilistic assessments rather than definitive truth judgments, reducing the risk of overreliance. Transparency is reinforced by presenting explanations and links to original sources, allowing users to independently review supporting evidence. Responsible AI deployment in this context demands a balance between automation and human agency.

Feasibility

The feasibility of this project is supported by the maturity of reverse image search technology and the demonstrated capabilities of multimodal AI models. The prototype implementation using Google Lens and Gemini confirms that real-time contextual image verification is technically achievable. Cloud infrastructure enables scalable deployment, while caching strategies mitigate excessive API costs. The integration of existing technologies reduces the need for extensive foundational model training, allowing development to focus on system integration and user experience at a reasonable cost.

Furthermore, the system is designed with flexibility in mind. The browser extension architecture allows for seamless integration with alternative vendors for image search and contextual analysis, ensuring adaptability to changing technological or cost constraints. This modular design enhances long-term feasibility and maintainability.

Budget

Table 1: *Estimated Monthly Operational Costs for Verilens Prototype*

Service	Cost/Mo.	Notes
Google Lens	$40.00	Includes 10,000 searches
Gemini 2.5 Flash	$11.25	Token-based pricing
TOTAL	**$51.25**	**~$0.51 per user / month**

With an estimation of two searches per day over a month, the approximate cost of the API calls used in the browser extension is outlined in Table 1. These estimates demonstrate that the operational expenses remain relatively low for individual users.

For a growing user base, scalability can be achieved by upgrading API pricing tiers, which is a straightforward process. Additionally, the use of caching mechanisms reduces redundant requests, further controlling operational costs. Overall, the browser extension maintains a cost structure that is both flexible and sustainable, making it viable for deployment to a reasonable number of users.

References

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